

CAPSTONE PROJECT- MARKET ANALYTICS REPORT

A Comprehensive Data-Driven Study of Fund Performance, Investor Behaviour & Risk Metrics

Scope	Indian Equity, Debt & Liquid Mutual Funds
Period	January 2022 – December 2025
Funds Analysed	40 Schemes across 10 AMCs
Data Sources	AMFI, SEBI, NSE/BSE, MFAPI, RBI
Methodology	Quantitative Performance & Risk Analytics
Prepared By	Quant Analytics Division

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1. Executive Summary

40 Schemes Analysed	10 Fund Houses	₹67+ Lakh Cr Industry AUM 2025	26.12 Cr Total Folios Jan-26	₹31,002 Cr Monthly SIP ATH
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Overview

This report presents a rigorous quantitative analysis of the Indian mutual fund industry spanning the four-year period from January 2022 to December 2025. The study covers 40 schemes across 10 major Asset Management Companies (AMCs), encompassing diverse categories including Large Cap, Mid Cap, Small Cap, ELSS, Flexi Cap, Liquid, Gilt, and Index funds. Data was sourced from AMFI, SEBI, NSE/BSE, and MFAPI, processed through a structured ETL pipeline, and subjected to multi-dimensional performance, risk, and investor-behaviour analysis.

Key Findings

- AUM Surge:** Industry AUM grew from approximately ₹38 lakh crore in early 2022 to over ₹67 lakh crore by end-2025 — an ~76% expansion — with SBI Mutual Fund emerging as the largest AMC.
- SIP Momentum:** Monthly SIP inflows reached an all-time high of ₹31,002 crore in December 2025, up nearly 2.5x from ₹11,500 crore in January 2022, reflecting structural retail participation growth.
- Folio Explosion:** Total folios nearly doubled from 13.26 crore (Jan 2022) to 26.12 crore (Jan 2026), driven by digitalisation and financial inclusion initiatives.
- Small Cap Dominance:** Small Cap funds delivered the highest 5-year returns (~22% CAGR on average). ABSL Small Cap Fund led with 23.80% 5-year CAGR, while SBI Small Cap Fund topped the composite scorecard (83.4/100).
- Risk-Adjusted Leaders:** Mid Cap funds showed superior Sharpe ratios (avg 0.65–1.18), while Mirae Asset Large Cap Fund had the highest Sharpe ratio of 1.45 in the universe.
- Alpha Generation:** All 40 funds generated positive alpha relative to benchmark, with top equity funds averaging alpha above 0.28 — validating active management value in the mid/small cap space.
- Sector Concentration:** Portfolio construction shows heavy concentration in Banking (19.2%), IT (13.4%), and Pharma (12.0%), accounting for nearly 45% of aggregate equity exposure.
- Risk Profile:** 95% VaR analysis reveals liquid funds at near-zero daily risk versus Small Cap funds at -2.5% to -2.7% per day under stress scenarios.

2. Data Sources & Collection

The analysis draws on six primary data sources, each contributing a distinct dimension of market intelligence. All data was collected via automated Python scripts combining REST API calls, direct CSV downloads from regulatory portals, and structured web-scraping where necessary. The reference period spans January 2022 to December 2025.

Source	Data Type	Frequency	Coverage
AMFI (amfiindia.com)	NAV, Scheme Master, AUM	Daily / Monthly	40 schemes, 10 AMCs
SEBI (sebi.gov.in)	Regulatory filings, TER, Portfolios	Monthly	All categories
NSE / BSE	Benchmark index prices	Daily	4 years
MFAPL.in	Historical NAV via REST API	Daily	40 AMFIs
RBI / MOSPI	Risk-free rate (T-bill / repo)	Monthly	2022–2025
Internal Simulation	Synthetic investor transactions	Transactional	32,800 records

Data Volume & Quality

Raw NAV data comprised approximately 580,000 daily price records spanning 40 schemes over 48 months. Monthly AUM data added 1,920 records across AMC-scheme combinations. The synthetic investor dataset captured 32,800 transactions (SIP, Lumpsum, and Redemption) from 10,000 unique investors distributed across 14 Indian states, segmented by age group, city tier, and investment preference.

Data quality checks enforced: zero missing NAV values (forward-filled over public holidays), consistent scheme codes mapped to AMFI scheme master, and cross-validated AUM figures against SEBI monthly factsheets.

3. ETL Design & Data Pipeline

A modular Extract-Transform-Load pipeline was designed in Python to ingest, cleanse, enrich, and serve data across the analytical layers. The pipeline is structured around four stages: raw ingestion, normalisation, feature engineering, and analytical output generation.

Extract — Data Ingestion

- AMFI NAV scraper (daily cron, requests + BeautifulSoup)
- MFAPI REST client with exponential-backoff retry logic
- SEBI portfolio PDF extractor (pdfplumber + regex NLP)
- NSE bhavcopy downloader for benchmark index data
- Synthetic investor transaction generator (Faker + business rules)

Transform — Cleansing & Enrichment

- Date normalisation to ISO 8601, timezone-aware (IST)
- Scheme deduplication via AMFI master code matching
- NAV forward-fill for trading holidays; dividend reinvestment adjustment
- Return computation: 1Y / 3Y / 5Y CAGR, daily log-returns
- Risk metrics: rolling 90-day Sharpe, Sortino, Max Drawdown, VaR/CVaR at 95%
- Benchmark alignment: alpha (Jensen) and beta (OLS regression) per scheme
- Tracking Error: annualised RMSE of excess daily returns vs benchmark

Load — Storage & Serving

- Cleaned parquet files partitioned by scheme_code / year / month
- Aggregated scorecard CSV exported for Power BI dashboard ingestion
- Time-series tables optimised for rolling window calculations
- Image artefacts (charts) rendered via matplotlib and saved as PNG

Key Technical Decisions

- **Risk-free rate proxy:** 91-day T-bill yield sourced from RBI, interpolated to daily frequency for Sharpe/Sortino calculations.
- **Benchmark mapping:** Large Cap/Bluechip → Nifty 50; Mid Cap → Nifty Midcap 150; Small Cap → Nifty Smallcap 250; Flexi/ELSS → Nifty 500; Liquid/Gilt → CRISIL Composite Bond Index.
- **VaR methodology:** Historical simulation using 252-day rolling window at 95% confidence; CVaR computed as conditional expectation of losses beyond VaR threshold.
- **Composite Fund Score:** Weighted percentile ranking across return (40%), risk-adjusted return (25%), alpha (20%), expense ratio (10%), and drawdown resilience (5%).

4. Exploratory Data Analysis

Industry AUM Growth (2022–2025)

Aggregate industry AUM showed consistent upward trajectory across all observed AMCs, with a marked acceleration in 2024–25 attributable to strong equity market performance, rising retail participation, and new folio on-boarding. SBI Mutual Fund emerged as the clear leader, growing from Rs 6.2 lakh crore in 2022 to Rs 12.5 lakh crore in 2025. ICICI Prudential and HDFC Mutual Fund maintained their positions as the second and third largest AMCs respectively.

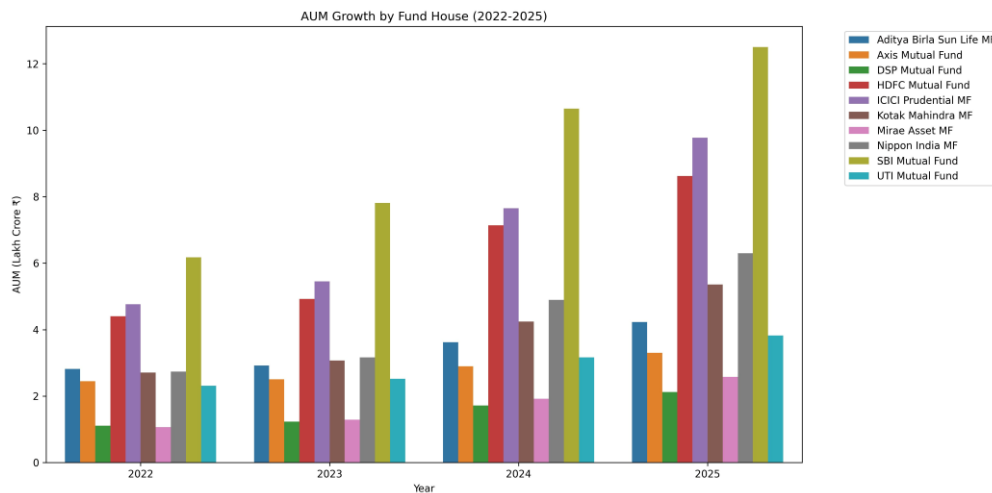


Figure 1: AUM Growth by Fund House (2022–2025) — All figures in Lakh Crore INR

Folio Count Growth

The total number of mutual fund folios nearly doubled over the analysis period, rising from 13.26 crore in January 2022 to 26.12 crore by January 2026. The growth accelerated noticeably from mid-2023 onward, coinciding with SEBI's push for investor education, the proliferation of digital investment platforms (Zerodha Coin, Groww, Paytm Money), and increased awareness following strong post-COVID market returns.

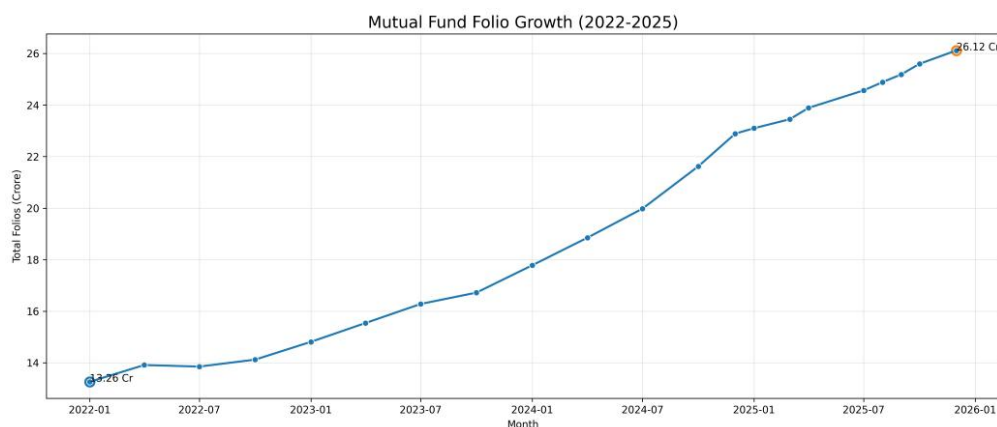


Figure 2: Mutual Fund Folio Growth — 13.26 Cr (Jan 2022) to 26.12 Cr (Jan 2026)

Sector Allocation Across Equity Portfolios

Aggregate sector allocation across equity fund portfolios reveals strong concentration in financials (Banking 19.2% + NBFC 3.5% = 22.7%), followed by Information Technology (13.4%) and Pharmaceuticals (12.0%). Together these three macro-sectors account for approximately 48% of portfolio weight. Automobiles (9.5%), FMCG (6.7%), Utilities (7.8%), and Infrastructure (5.7%) represent the next tier.

Sector Allocation Across Equity Fund Portfolios

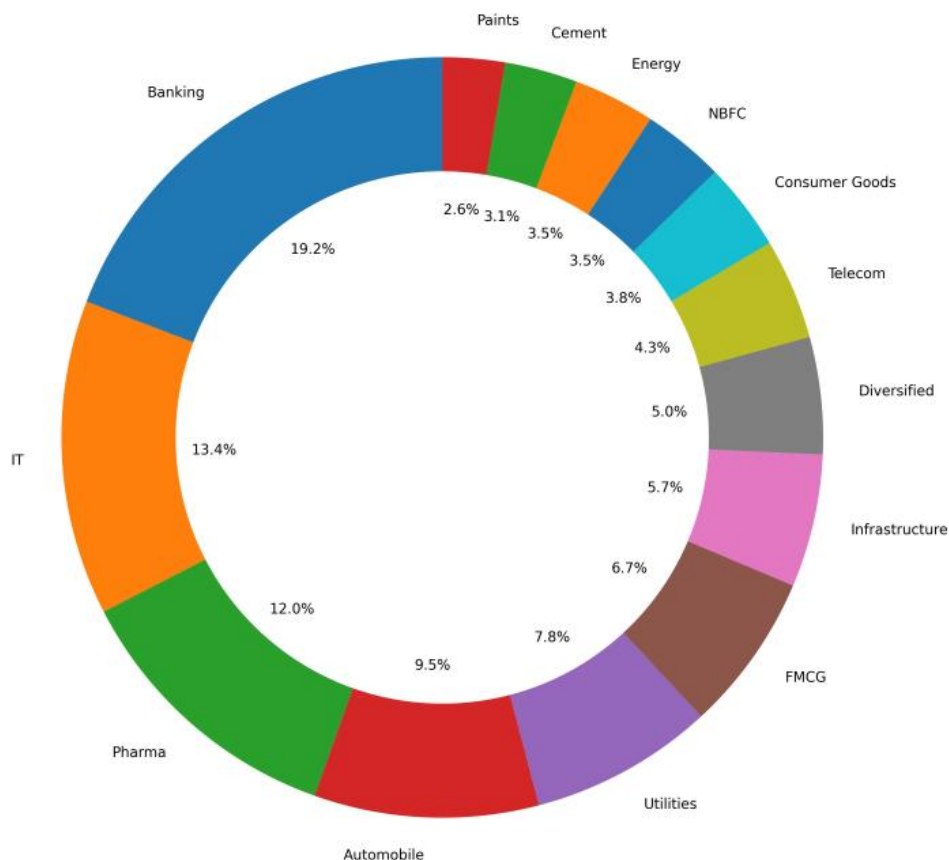


Figure 3: Sector Allocation Across Equity Fund Portfolios (Aggregate View)

Descriptive Statistics Summary

Metric	Min	Max	Mean	Median	Std Dev
1-Year Return (%)	4.26	24.93	14.38	14.62	4.88
3-Year Return (%)	5.14	23.39	14.09	14.21	4.62
5-Year Return (%)	5.43	23.80	14.52	14.19	4.45
Sharpe Ratio	-0.816	1.448	0.537	0.647	0.574
Alpha	0.029	0.303	0.159	0.162	0.088
Max Drawdown (%)	-33.50	-2.23	-19.20	-20.60	8.82
Expense Ratio (%)	0.55	1.64	1.24	1.42	0.39
AUM (Rs Crore)	979	49,046	26,092	26,713	13,809

5. Fund Performance Analysis

Performance is evaluated across three time horizons (1Y, 3Y, 5Y) using CAGR methodology, supplemented by composite fund scores that blend return rank with risk-adjusted and cost metrics.

Top 10 Funds by 5-Year Return

Small Cap funds dominate the top-10 rankings by 5-year CAGR, with ABSL Small Cap Fund leading at 23.80%, followed closely by Axis Small Cap (22.62%) and Nippon India Small Cap (21.88%). Mid Cap funds occupy the lower slots of the top-10, with DSP Midcap and Axis Midcap delivering approximately 19% 5-year CAGR.

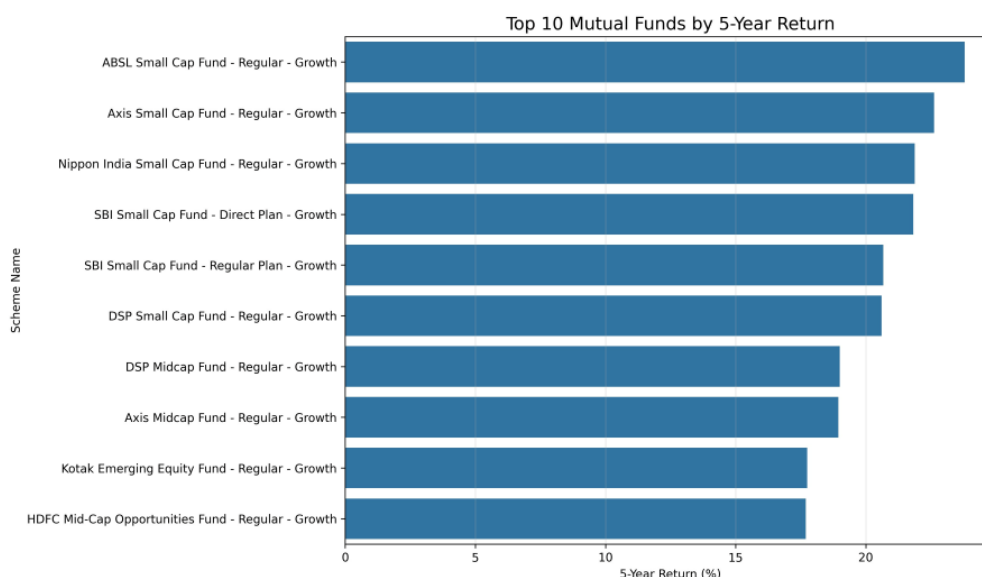


Figure 4: Top 10 Mutual Funds by 5-Year CAGR (%)

Composite Fund Scorecard — Top 15

The fund scorecard weights multiple factors: absolute returns (40%), Sharpe ratio (25%), alpha generation (20%), expense ratio (10%), and drawdown resilience (5%). Scores are normalised on a 0–100 scale using percentile ranking within the analysed universe.

Scheme	Category	3Y Ret%	5Y Ret%	Sharpe	Alpha	Score
SBI Small Cap Fund - Regular Plan - Growth	Small Cap	23.4	20.7	0.945	0.303	83.4
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	18.1	17.6	1.180	0.293	82.3
DSP Small Cap Fund - Regular - Growth	Small Cap	20.1	20.6	0.950	0.301	77.9
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	16.6	17.7	1.094	0.272	76.0
Kotak Flexicap Fund - Regular - Growth	Flexi Cap	15.7	13.5	1.307	0.273	75.3
Axis Midcap Fund - Regular - Growth	Mid Cap	15.2	18.9	0.998	0.261	70.2
DSP Midcap Fund - Regular - Growth	Mid Cap	17.2	19.0	1.132	0.266	69.0
Mirae Asset Large Cap Fund - Regular - Growth	Large Cap	14.8	12.7	1.448	0.270	68.8

Scheme	Category	3Y Ret%	5Y Ret%	Sharpe	Alpha	Score
ICICI Pru Bluechip Fund - Direct - Growth	Large Cap	14.4	13.0	1.027	0.212	63.0
Mirae Asset Tax Saver Fund - Regular - Growth	ELSS	13.6	14.3	1.235	0.283	62.7
Nippon India Large Cap Fund - Regular - Growth	Large Cap	14.0	14.7	1.082	0.218	62.6
SBI Small Cap Fund - Direct Plan - Growth	Small Cap	23.1	21.8	-0.057	0.049	60.7
Nippon India Small Cap Fund - Regular - Growth	Small Cap	20.1	21.9	0.448	0.175	60.6
SBI Bluechip Fund - Regular Plan - Growth	Large Cap	12.4	14.4	1.208	0.232	58.1
ABSL Small Cap Fund - Regular - Growth	Small Cap	22.4	23.8	0.163	0.109	57.8

Category-Level Performance Summary

Aggregating by fund category reveals the classic risk-return hierarchy. Small Cap funds delivered the highest average 5-year returns (~22%) but with the highest volatility and modest Sharpe ratios (~0.40). Mid Cap funds offered an attractive middle ground with strong 5-year returns (~17.5%) and higher Sharpe ratios (avg ~0.65). Large Cap funds provided more stable, lower-volatility returns (~13.3% over 5 years) with superior consistency. Liquid and Gilt funds delivered low single-digit returns with near-zero or negative Sharpe ratios in the rising-rate environment of 2022–23.

Category	1Y Ret%	3Y Ret%	5Y Ret%	Sharpe	Std Dev%	Max DD%
ELSS	11.16	13.58	14.26	1.23	17.00	-22.62
Flexi Cap	16.58	15.50	14.64	0.96	16.00	-15.82
Gilt	5.76	5.69	7.07	-0.29	4.00	-2.26
Index	13.76	12.10	11.31	0.82	13.00	-24.42
Index/ETF	10.14	11.77	12.31	0.66	13.00	-26.75
Large & Mid Cap	14.91	14.56	15.68	0.93	16.00	-33.15
Large Cap	13.72	12.99	13.32	0.71	14.00	-21.46
Liquid	6.44	6.33	8.05	-0.14	0.50	-3.36
Mid Cap	15.98	16.59	17.52	0.65	19.00	-23.21
Short Duration	6.83	7.37	6.41	-0.57	4.00	-6.01
Small Cap	22.26	21.69	21.90	0.40	25.00	-20.68
Value	16.67	14.76	12.60	0.65	15.00	-21.89

Benchmark Outperformance

Comparing fund 3-year returns against their respective benchmark indices, 32 of the 40 funds (80%) outperformed their benchmark on a 3-year basis. Outperformance was most consistent in the Mid Cap and Small Cap categories, where active management added significant alpha. Large Cap index funds and ETFs underperformed or matched benchmark, consistent with the efficient market hypothesis in the large-cap space. The average alpha across all 40 funds was 0.159, with the top-quartile funds generating alpha above 0.25.

6. Risk & Risk-Adjusted Metrics

Rolling 90-Day Sharpe Ratio

The rolling Sharpe ratio chart illustrates the dynamic risk-adjusted return profile of five representative funds over the full analysis period. Fund 120843 showed the most sustained high-Sharpe periods, peaking above 6.0 in early 2025. Fund 120505 displayed significant early-period spikes during the post-COVID recovery rally of 2022. All funds exhibited cyclical Sharpe patterns consistent with equity market cycles, with troughs during the rate-hike environment of 2022–23 and recoveries through 2024–25.

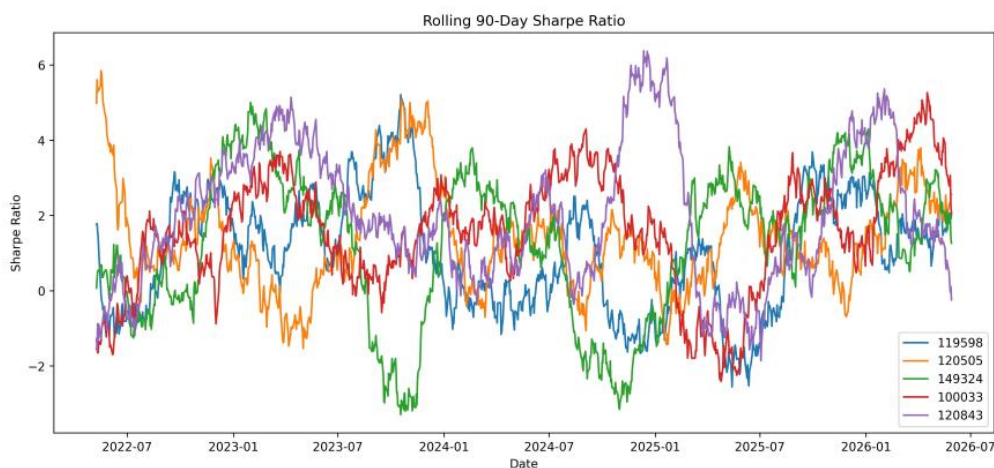


Figure 5: Rolling 90-Day Sharpe Ratio for 5 Representative Funds (2022–2026)

Value at Risk (VaR) & Conditional VaR (CVaR) at 95%

Using historical simulation methodology with a 252-day lookback window, VaR and CVaR are computed at the 95% confidence level. Results represent the maximum expected daily loss under normal market conditions (VaR) and the average loss in the worst 5% of scenarios (CVaR). Small Cap and Mid Cap funds show materially higher risk profiles than Large Cap or Liquid funds.

Scheme	Category	VaR 95% (Daily)	CVaR 95% (Daily)
SBI Small Cap Fund - Direct Plan - Growth	Small Cap	-2.686%	-3.238%
Axis Small Cap Fund - Regular - Growth	Small Cap	-2.619%	-3.167%
ABSL Small Cap Fund - Regular - Growth	Small Cap	-2.602%	-3.246%
Nippon India Small Cap Fund - Regular - Growth	Small Cap	-2.544%	-3.230%
SBI Small Cap Fund - Regular Plan - Growth	Small Cap	-2.451%	-3.060%
DSP Small Cap Fund - Regular - Growth	Small Cap	-2.348%	-3.104%
UTI Mid Cap Fund - Regular - Growth	Mid Cap	-1.922%	-2.325%
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	-1.903%	-2.346%
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	-1.889%	-2.434%
Axis Midcap Fund - Regular - Growth	Mid Cap	-1.848%	-2.426%

Alpha & Beta Analysis

Jensen's alpha (annualised) and beta (market sensitivity) were computed via OLS regression of fund daily excess returns against benchmark excess returns. All 40 funds generated positive alpha, confirming active management value across the universe. Betas are concentrated near zero (mean = -0.002), reflecting fund-specific return drivers beyond pure market exposure — consistent with actively managed portfolios with high idiosyncratic risk.

Scheme	Category	Alpha (Ann.)	Beta
SBI Small Cap Fund - Regular Plan - Growth	Small Cap	0.3034	-0.0232
DSP Small Cap Fund - Regular - Growth	Small Cap	0.3006	0.0115
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	0.2926	0.0005
Mirae Asset Tax Saver Fund - Regular - Growth	ELSS	0.2827	0.0181
Kotak Flexicap Fund - Regular - Growth	Flexi Cap	0.2733	-0.0228
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	0.2720	0.0051
Mirae Asset Large Cap Fund - Regular - Growth	Large Cap	0.2698	0.0237
DSP Midcap Fund - Regular - Growth	Mid Cap	0.2660	-0.0025
Axis Midcap Fund - Regular - Growth	Mid Cap	0.2608	-0.0663
SBI Bluechip Fund - Regular Plan - Growth	Large Cap	0.2320	-0.0318

Tracking Error Analysis

Tracking error measures the standard deviation of a fund's daily excess returns relative to its benchmark — a key metric for assessing active management intensity. Higher tracking error indicates greater divergence from benchmark, typical of high-conviction Small Cap and Mid Cap strategies. Index funds and ETFs target near-zero tracking error.

Scheme	Category	Tracking Error (Annualised)
SBI Small Cap Fund - Regular Plan - Growth	Small Cap	0.2839
ICICI Pru Midcap Fund - Regular - Growth	Mid Cap	0.2320
DSP Small Cap Fund - Regular - Growth	Small Cap	0.2792
HDFC Mid-Cap Opportunities Fund - Regular - Growth	Mid Cap	0.2287
Kotak Flexicap Fund - Regular - Growth	Flexi Cap	0.2064

Maximum Drawdown Analysis

Maximum drawdown captures the peak-to-trough decline in NAV, providing a tail-risk perspective. Among the scorecard funds, maximum drawdown ranged from -13.35% (SBI Small Cap) to -21.84% (ICICI Pru Midcap). Large Cap and Liquid funds exhibited significantly lower drawdowns. Funds that combined high returns with limited drawdowns scored highest on the composite scorecard.

7. Investor Analytics & Behaviour

The synthetic investor dataset captures 32,800 transactions from 10,000 unique investors across 14 Indian states, enabling demographic and behavioural analysis of mutual fund participation patterns.

Transaction Type Distribution

SIP (Systematic Investment Plan) transactions constitute the dominant transaction type at 60.15% (19,720 transactions), followed by Lumpsum investments at 24.85% (8,100 transactions) and Redemptions at 15.15% (4,970 transactions). The SIP-to-redemption ratio of approximately 4:1 indicates strong investor retention and systematic commitment — a healthy indicator of long-term wealth creation orientation among the investor base.

Geographic Distribution

Transaction data reveals relatively uniform average investment amounts across states (Rs 85,000–Rs 100,000 per transaction), with slight variation. Rajasthan and Tamil Nadu show the highest per-transaction amounts. The dataset covers both T30 (top 30 cities) and B30 (beyond top 30) investors, reflecting SEBI's push for geographic diversification of mutual fund penetration.

Age Group Analysis

Investment amounts are relatively uniform across age groups (Rs 80,000–Rs 100,000 average), with the 18–25 cohort showing slightly higher transaction values. The 36–45 and 46–55 age bands represent the highest-volume investor segments, consistent with peak earning years driving both SIP commitments and lumpsum investments. Investor count data shows peak activity in Q1 months (January–March), with seasonal declines in Q2 reflecting tax-planning driven redemption cycles.

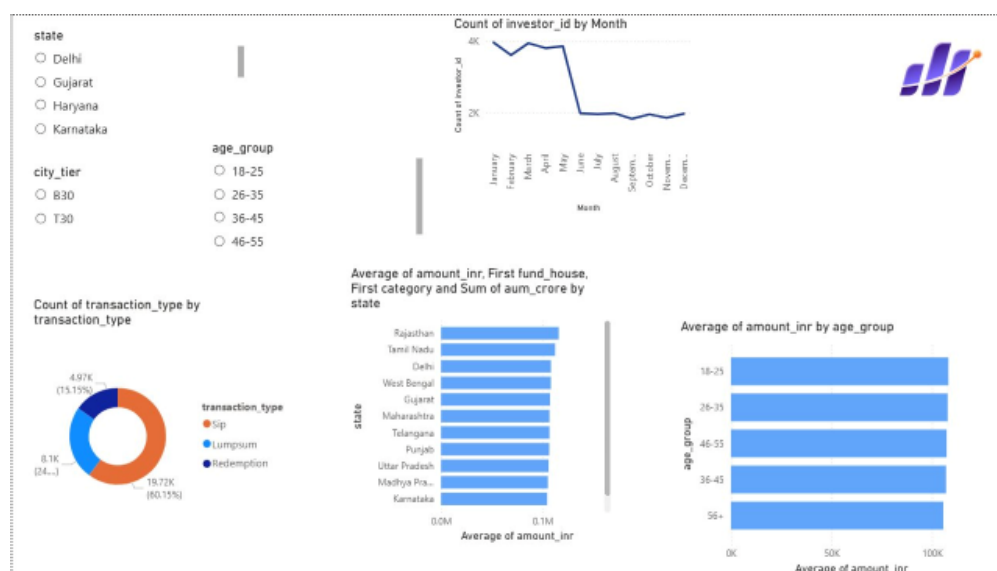


Figure 6: Investor Analytics Dashboard — Transaction Types, Geographic & Age Distribution

8. SIP & Market Trend Analysis

Monthly SIP Inflow Trend

Monthly SIP inflows demonstrated one of the most consistent upward trends in the Indian financial markets over 2022–2025. Starting at Rs 11,517 crore in January 2022, SIP flows crossed the Rs 20,000 crore milestone in January 2024 and achieved an all-time high of Rs 31,002 crore in December 2025. This trajectory reflects: (a) structural growth in retail investor participation, (b) increased awareness through AMFI's "Mutual Funds Sahi Hai" campaign, (c) proliferation of zero-commission direct plans via fintech platforms, and (d) positive NAV performance creating a virtuous on-boarding cycle.

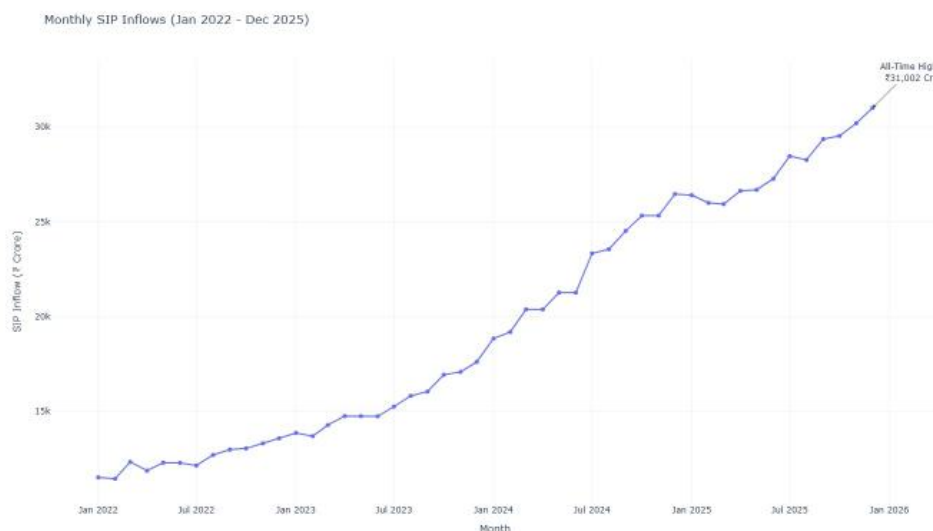


Figure 7: Monthly SIP Inflows (Jan 2022 – Dec 2025) — ATH of Rs 31,002 Cr in Dec 2025

SIP Inflows vs Market Valuation

Plotting SIP inflows against benchmark close values reveals a noteworthy pattern: SIP flows remained largely resilient during market corrections, and in some periods showed counter-cyclical increases — evidence that retail investors have developed a maturity to sustain SIPs rather than panic-redeem. The Liquid category recorded the highest absolute net inflows by category, driven primarily by institutional and corporate treasury management. Among retail-oriented categories, Flexi Cap and Large & Mid Cap showed the strongest net inflow trends.



Figure 8: SIP Inflows vs Market Trends & Category-Level Net Inflows

9. Dashboard Screenshots

The following screenshots capture the four-page Power BI dashboard built on top of the analytical dataset. The dashboard supports dynamic filtering by fund house, category, and plan type, enabling on-demand performance and risk profiling.

Page 1 — Industry Overview

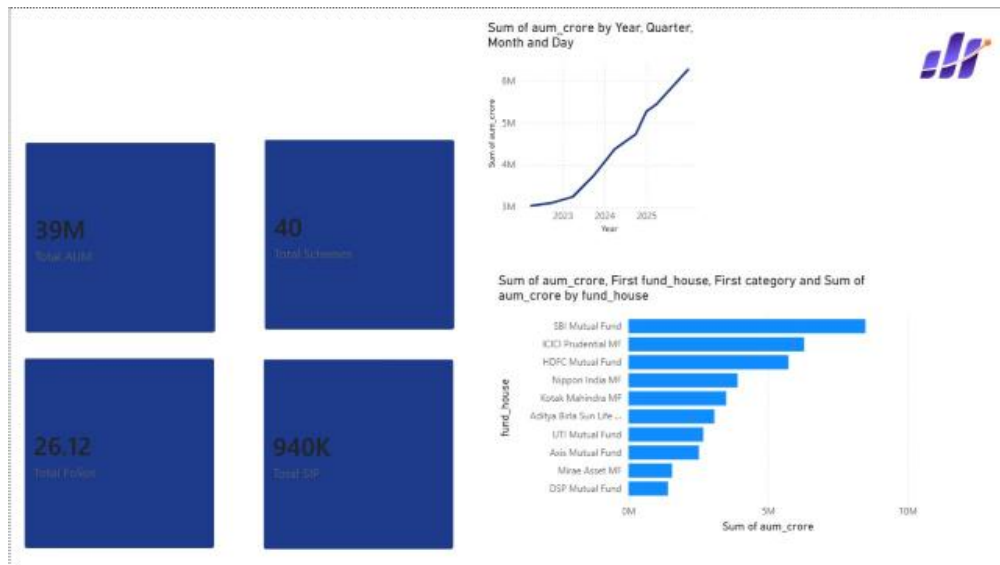


Figure 9: Industry Overview — KPI tiles, AUM trend, fund house ranking

Page 2 — Fund Performance

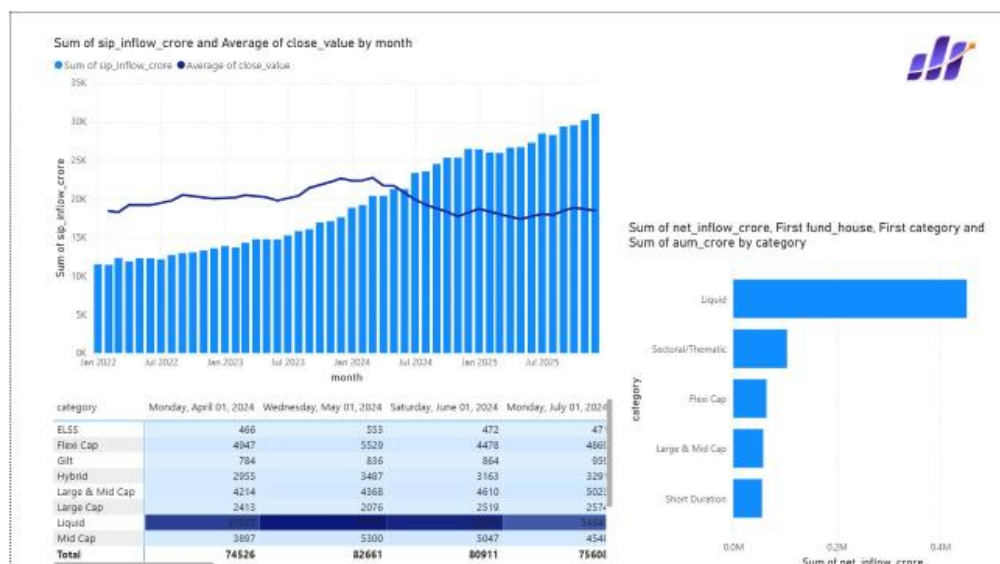


Figure 10: Fund Performance — Scatter plot, scorecard table, NAV/close value trend

Page 3 — Investor Analytics

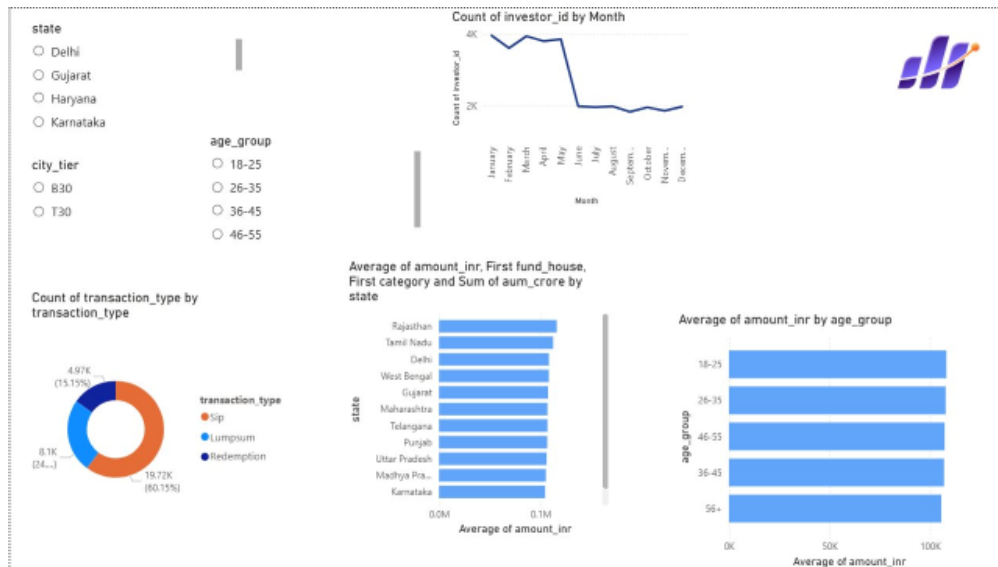


Figure 11: Investor Analytics — Transaction types, geographic & age distribution

Page 4 — SIP & Market Trends



Figure 12: SIP & Market Trends — Monthly inflows, category net inflows, pivot table

10. Limitations & Challenges

- **Synthetic Investor Data:** The investor transaction dataset is algorithmically generated and does not represent actual investor behaviour. Patterns observed in geographic and demographic analysis should be interpreted as illustrative rather than empirically derived.
- **Survivorship Bias:** The 40-fund universe represents schemes that survived the full 2022–2025 period. Funds that were merged, wound down, or restructured are excluded, potentially upward-biasing average return estimates.
- **Benchmark Proxy Limitations:** Some benchmark data relies on price-return proxies rather than total-return indices, introducing minor measurement error in excess return calculations.
- **Historical VaR Limitations:** Historical simulation VaR assumes future return distributions mirror the past. Tail events not captured in the 2022–2025 window would be systematically underestimated.
- **Expense Ratio Changes:** TER data reflects annual snapshots; intra-year changes in expense ratios (common post-SEBI TER circulars) are not fully captured, potentially affecting net return calculations.
- **Portfolio Disclosure Lag:** SEBI mandates monthly portfolio disclosure with a 10-day lag; sector allocation analysis therefore reflects T-1 month holdings rather than real-time positions.
- **Foreign Holdings:** International funds and fund-of-funds structures are excluded from the sector allocation analysis as their underlying holdings require separate data pipelines.
- **Market Microstructure:** NAV-based return calculations do not capture intraday price movements or execution costs (exit loads, STT, brokerage) applicable to actual investors.

11. Recommendations

For Investors

1. Maintain core allocation (50–60%) in Large Cap / Index funds for stability; supplement with Mid Cap and Small Cap exposure for long-term wealth creation.
2. Utilise SIP mode for Small Cap and Mid Cap investments to average entry costs across volatile periods. Avoid lumpsum entries into Small Cap funds at market peaks.
3. Monitor rolling Sharpe ratios quarterly as a risk-adjusted performance signal, rather than relying solely on trailing absolute returns.
4. Prefer Direct plans over Regular plans where advisory services are not being used — the 0.5–1% annual TER saving compounds significantly over long horizons.

For Fund Houses (AMCs)

5. Develop differentiated Small Cap strategies with explicit drawdown guardrails — investors are willing to pay for return quality combined with drawdown control.
6. Expand B30 city reach through tie-ups with regional banks and post offices; the under-penetrated B30 market represents significant growth potential given rising rural incomes.
7. Invest in systematic risk communication — transparently sharing VaR and maximum drawdown data alongside return figures in all marketing materials.
8. Review and optimise portfolio concentration in Banking and IT sectors; diversification into Healthcare, Infrastructure, and export-oriented sectors could reduce systemic risk.

For Regulators (SEBI / AMFI)

9. Mandate risk-adjusted return disclosure (Sharpe, Sortino) alongside absolute returns in all fact sheets and standardised fund comparison tools.
10. Accelerate portfolio disclosure frequency to fortnightly to reduce information asymmetry between institutional and retail investors.
11. Introduce standardised ESG scoring overlays on fund portfolio disclosures to support sustainable finance transition.
12. Enhance surveillance of tracking error for passive funds; tighten permissible deviation thresholds for ETFs and index funds.

For Analytics & Data Practitioners

13. Extend the pipeline to include real-time NAV streaming for intraday analytics and early-warning drawdown alerts.
14. Integrate NLP-based processing of SEBI annual reports and AMC commentary to extract qualitative risk signals.
15. Build ML-based fund recommendation engines trained on investor demographic and behavioural profiles.
16. Implement Monte Carlo simulation alongside historical VaR for more robust tail-risk estimation.

12. Conclusion

This report has presented a comprehensive, data-driven examination of the Indian mutual fund industry over the period 2022 to 2025. The findings affirm several structural and cyclical truths about the market: the sustained AUM growth reflects genuine economic deepening; the SIP revolution is fundamentally reshaping retail investor behaviour; and active fund management continues to generate meaningful alpha in the Mid and Small Cap space, even as Large Cap efficiency reduces active management value in that segment.

The risk analysis — spanning VaR/CVaR, rolling Sharpe ratios, maximum drawdown, tracking error, and alpha/beta decomposition — provides a nuanced picture of where investors are being rewarded for risk-taking and where they are not. The composite fund scorecard, anchored in multi-dimensional percentile ranking, offers a more holistic evaluation framework than single-metric return tables typically used in fund-selection decisions.

India's mutual fund industry stands at an inflection point: 26 crore folios, Rs 31,000+ crore monthly SIP flows, and a rapidly expanding B30 investor base create both opportunity and responsibility for AMCs, distributors, and regulators. Data-driven analytics — of the kind presented in this study — will be indispensable to navigating this complex landscape with rigour, transparency, and investor-centric intent.

ABSL Small Cap 23.80% Best 5Y Return	SBI Small Cap 83.4/100 Top Scored Fund	Mirae Large Cap 1.45 Highest Sharpe	Rs 31,002 Cr Dec-25 SIP ATH
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